	Raffles Institution Physics Department Revision Package 1	**Nuclear Physics**	**20**

Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs

1. **[AJC/H2/Prelims 2010/P1/39]**
When a beam of alpha particles is directed at a piece of thin gold foil, most of the alpha particles pass through undeflected. We can deduce that

A the mass of a gold atom is concentrated in the small gold nuclei.

B the gold atoms are very small.

C there are large empty spaces in the gold atoms.

D the positive charges are concentrated in small gold nuclei.

2. **[AJC/H2/Prelims 2010/P1/40]**
Which statement about nuclear fission is correct?

A It is random and spontaneous.

B The parent nuclide emits a neutron and decay into a different element.

C Uranium has lower binding energy than each of its fission products.

D It produces less massive elements with higher binding energy per nucleon.

3. **[CJC/H2/Prelims 2010/P1/39]**
In an experiment to investigate the nature of the atom, a very thin gold film was bombarded with α-particles.
Which pattern of deflection of the α-particles was observed?

A A few α-particles were deflected through angles greater than a right angle.

B All α-particles were deflected from their original path.

C No α-particle was deflected through an angle greater than a right angle.

D An interference pattern was observed.

4. **[CJC/H2/Prelims 2010/P1/40]**
A sample of pure potassium chloride is found to be radioactive due to the presence of ^{40}K. The sample contains 9.49×10^{19} atoms of ^{40}K when the activity is measured to be 1600 Bq.
What is the half-life of the radioactive decay of ^{40}K?

A 1.67×10^{-17} s B 5.93×10^{16} s C 4.11×10^{16} s D 1.30 years

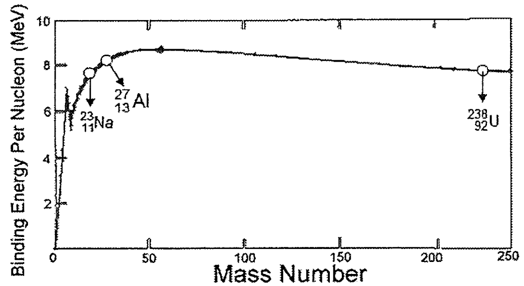
5. **[Dunman High/H2/Prelims 2010/P1/39]**
One reaction that may be used for nuclear fusion is shown

$$^{7}_{3}\text{Li} + ^{2}_{1}\text{H} \rightarrow 2\left(^{4}_{2}\text{He}\right) + \text{X}$$

What is particle X?

A an α-particle B an electron C a neutron D a proton

6. **[Dunman High/H2/Prelims 2010/P1/40]**
The diagram below shows a graph of the binding energy per nucleon for a number of naturally occurring nuclides plotted against their mass number.



Which statement is a correct deduction from the graph?

A Uranium (^{238}U) is the most stable nuclides amongst the 3 plotted.

B Energy will be released if sodium (^{23}Na) undergoes nuclear fission.

C Aluminum (^{27}*Al*) will spontaneously decay to sodium (^{23}Na) with alpha particle.

D Nuclear fusion between sodium (^{23}Na) and aluminum (^{27}*Al*) will release energy.

7. **[HCI/H2/Prelims 2010/P1/38]**
It is possible for an electron to annihilate a positron to produce two identical photons. The mass m of the positron is equal to the mass of the electron.
What is the minimum wavelength λ of the photons produced?

A $\dfrac{h}{mc}$ B $\dfrac{h}{mc^2}$ C $\dfrac{h}{2mc}$ D $\dfrac{2h}{mc^2}$

8. **[HCI/H2/Prelims 2010/P1/39]**
A sample of a radioactive material contains 10^{18} atoms. The half-life of the material is 2.00 days.
What is the activity of the sample after 5.00 days?

A 1.77×10^{5} disintegrations s^{-1}

B 7.09×10^{11} disintegrations s^{-1}

C 2.05×10^{12} disintegrations s^{-1}

D 6.13×10^{16} disintegrations s^{-1}

9. [HCI/H2/Prelims 2010/P1/40]

A radioactive source contains two materials. One has a half-life of 4 days and decays by the emission of alpha particles whilst the other has a half-life of 3 days and emits beta particles. The initial count rate is 160 Bq but when a sheet of paper is placed in between the source and the detector, the reading drops to 96 Bq.

What is the count rate after 12 days, without the paper present?

A 10 Bq B 14 Bq C 16 Bq D 20 Bq

10. [IJC/H2/Prelims 2010/P1/39]

When an isotope of boron, $^{10}_{5}B$ captures a slow neutron, it splits into a lithium $^{7}_{3}Li$ and an alpha particle. An emission of γ-ray occurs during this reaction. The nuclear binding energies are

$^{10}_{5}B$: 64.94 MeV

$^{7}_{3}Li$: 39.25 MeV

$^{4}_{2}He$: 28.48 MeV

The total kinetic energy of the products produced, $^{7}_{3}Li$ and $^{4}_{2}He$ is 2.31 MeV.

What is the energy of the γ-ray emitted?

A 0.48 MeV B 2.79 MeV C 10.77 MeV D 25.69 MeV

11. [IJC/H2/Prelims 2010/P1/40]

The half-life of a certain radioactive material is 3.0 s.

How long does it take for its activity to reduce by 90 %?

A 0.46 s B 5.4 s C 10 s D 11 s

12. [MI/H2/Prelims 2010/P1/39]

Which of the following equations correctly shows an α-particle causing a nuclear reaction?

A $^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}n$

B $^{17}_{8}O + ^{4}_{2}He \rightarrow ^{20}_{9}F + ^{1}_{1}p$

C $^{17}_{8}O + ^{0}_{-1}e \rightarrow ^{13}_{5}B + ^{4}_{2}He$

D $^{14}_{7}N + ^{1}_{1}p \rightarrow ^{11}_{6}C + ^{4}_{2}He$

13. [MI/H2/Prelims 2010/P1/40]

Initially, a source comprises N_0 nuclei of a radioactive nuclide.

What is the number of nuclei decayed after a time interval of three half-lives?

A $\frac{N_0}{16}$ B $\frac{N_0}{8}$ C $\frac{7N_0}{8}$ D $\frac{15N_0}{16}$

14. [MJC/H2/Prelims 2010/P1/39]

The radioactive isotope of iodine, ^{123}I, is often used to test for overall thyroid function in patients. The thyroid of an individual with hypothyroid condition will accumulate less iodine than that of a normal individual.

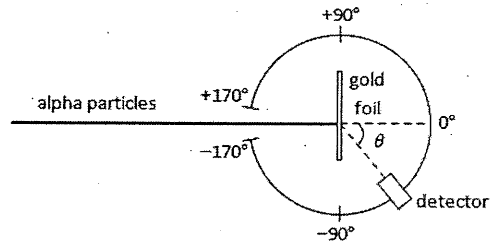
An initial dose of ^{123}I with an activity of 30 μCi was administered intravenously to a patient. Twenty-four hours after injecting the radiopharmaceutical, the activity emanating from the thyroid region is monitored and found to be 4 μCi.

What percentage of the injected ^{123}I was concentrated in the thyroid? (Half-life of ^{123}I = 13 hrs).

A 13% B 24% C 26% D 48%

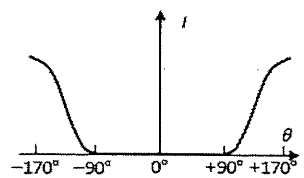
15. [MJC/H2/Prelims 2010/P1/40]

The figure below shows the apparatus used to repeat the alpha-particle scattering experiment. The detector measures the intensity of the alpha-radiation I at various angular positions θ.

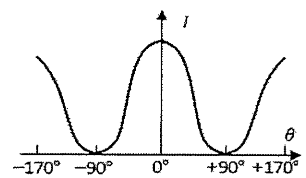


Which of the graphs best represents the variation of I with θ?

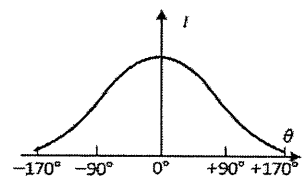
A



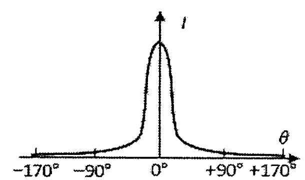
B



C



D



16. [NJC/H2/Prelims 2010/P1/39]

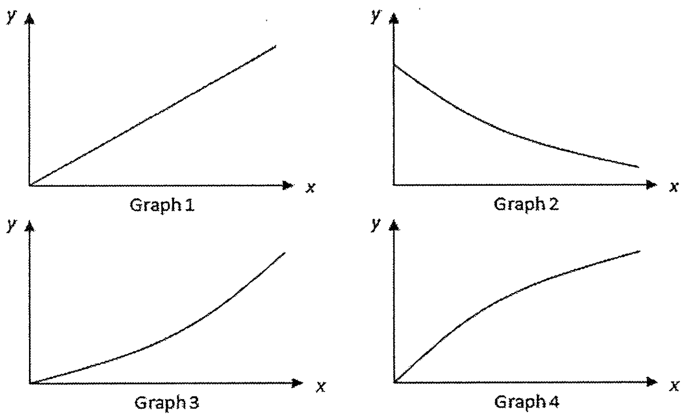
When a detector is pointed towards a radioactive sample that is 80.0 cm away and has a half-life of 20 minutes, it gives an average count-rate of 78 s^{-1}. In the absence of the source, the average count-rate is 10 s^{-1}.

What average count-rate is expected 40 minutes later, with the detector still pointed towards the sample but now located 40.0 cm away? Regard the sample to be a point source of radiation.

A 17 s^{-1} B 27 s^{-1} C 68 s^{-1} D 78 s^{-1}

17. [NJC/H2/Prelims 2010/P1/40]

The nucleus of an unstable element decays by α-particle emission into a stable daughter. The process is observed in an experiment.



Which row is correct?

	number of daughter nuclei vs time	rate of alpha emissions vs number of parent nuclei present
A	Graph 2	Graph 3
B	Graph 2	Graph 1
C	Graph 4	Graph 3
D	Graph 4	Graph 1

18. [NYJC/H2/Prelims 2010/P1/39]

The fission of a heavy nucleus gives, in general, two lighter product nuclei. One such reaction takes place when Uranium-235 nucleus undergoes fission by slow-moving neutron:

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{139}_{54}Xe + ^{95}_{38}Sr + 2^{1}_{0}n$$

Which of the following statements is true about the product nuclei produced?

A They have a total rest mass that is greater than that of the original nucleus.

B The sum of their mass defect is greater than that of the original nucleus.

C The sum of their kinetic energies is smaller than that of the original nucleus.

D The sum of their binding energies is less than that of the original nucleus.

19. [NYJC/H2/Prelims 2010/P1/40]

Radioactive ^{14}C dating was used to find the age of a wooden archeological specimen. Measurements were taken in three situations for which the following count rates were obtained:

specimen	count rate / counts per minute
1 g sample of living wood	80
1 g sample of archaeological specimen	35
no sample	25

The half-life of ^{14}C is known to be 5700 years.

What was the approximate age of the archeological specimen?

A 3700 years B 6800 years C 14000 years D 17000 years

20. [PJC/H2/Prelims 2010/P1/39]

The half-life of a certain radioactive isotope is 32 hours.

What fraction of a sample would remain after 16 hours?

A 0.25 B 0.29 C 0.71 D 0.75

21. [PJC/H2/Prelims 2010/P1/40]

A high energy α-particle collides with a $^{14}_{7}N$ nucleus to produce a $^{17}_{8}O$ nucleus.

What could be the other products of this collision?

A a γ-photon alone

B a γ-photon and a β-particle

C a γ-photon and a neutron

D a γ-photon and a proton

22. [RI/H2/Prelims 2010/P1/39]

Two samples of radioactive nuclides X and Y are prepared. Y has twice the initial activity and twice the half-life of X.

After 6 half-lives of X, what is the ratio of the activity of X to Y?

A $\frac{1}{2}$ B $\frac{1}{4}$ C $\frac{1}{8}$ D $\frac{1}{16}$

23. [RI/H2/Prelims 2010/P1/40]

The nuclear reaction P + Q $\rightleftharpoons$ X + Y proceeds with a release of energy.

Which statement must be correct?

A Mass of X and Y is larger than mass of P and Q.

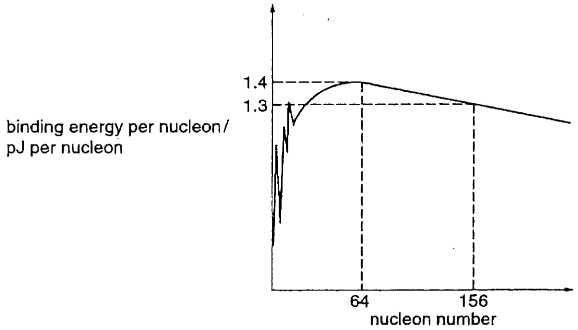
B Momentum of X and Y is larger than momentum of P and Q.

C Total binding energy of X and Y is larger than total binding energy of P and Q.

D Binding energy per nucleon of both X and Y are larger than binding energy per nucleon of P or Q.

24. [RVHS/H2/Prelims 2010/P1/39]

The graph shows how the binding energy per nucleon varies with the nucleon number for naturally occurring nuclides.



What is the total binding energy of the nuclide $^{156}_{64}Gd$?

A 1.3 pJ B 1.4 pJ C 90 pJ D 203 pJ

25. [RVHS/H2/Prelims 2010/P1/40]

A sample of a radioactive isotope of half-life $t_{1/2}$ initially contains N atoms.

Which one of the following gives the number of atoms of this isotope that have decayed after a time $3t_{1/2}$?

A $\frac{7}{8}N$ B $\frac{2}{3}N$ C $\frac{1}{3}N$ D $\frac{1}{8}N$

26. [SAJC/H2/Prelims 2010/P1/39]

Uranium-235 undergoes fission as shown in the equation below.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{143}_{55}Cs + ^{90}_{37}Rb + 3^{1}_{0}n$$

195 MeV of energy was released in the reaction. The binding energy per nucleon for uranium-235 is 7.6 MeV, and those for caesium and rubidium are approximately x MeV.

What is the value of x?

A 8.5 MeV B 9.7 MeV C 11.2 MeV D 13.3 MeV

27. [SAJC/H2/Prelims 2010/P1/40]

The number of undecayed radioactive nuclei, N, of cobalt $^{60}_{27}Co$ at three different timings are as follows:

t / years	N
0	500000
12	M
16	31250

What is the value of M?

A 25000 B 62500 C 100000 D 125000

28. [TJC/H2/Prelims 2010/P1/39]

The fusion of two deuterium nuclei produces a nuclide of helium plus a neutron and liberates 3.27 MeV of energy.

How does the mass of the two deuterium nuclei compare with the combined mass of the helium nucleus and neutron?

A It is 5.8×10^{-30} kg greater before fusion.

B It is 5.8×10^{-30} kg greater after fusion.

C It is 5.8×10^{-36} kg greater before fusion.

D It is 5.8×10^{-36} kg greater after fusion.

29. [TJC/H2/Prelims 2010/P1/40]

In an experiment on the transport of nutrients in the root structure of a plant, two radioactive nuclides X and Y were used.

Initially, the ratio $\frac{\text{number of nuclei of nuclide X}}{\text{number of nuclei of nuclide Y}}$ is 2.5.

Three days later, the ratio $\frac{\text{number of nuclei of nuclide X}}{\text{number of nuclei of nuclide Y}}$ is 5.0.

Nuclide Y has a half-life of 1.5 days.

What is the half-life of nuclide X?

A 2.0 days B 3.0 days C 4.5 days D 6.0 days

30. [VJC/H2/Prelims 2010/P1/39]

α-particles are deviated by thin metal foils through angles that ranges from 0° to 180°.

Which of the following is the explanation?

A Diffraction of α-particle by the crystal lattice

B Scattering of α-particle by small but heavy regions of positive charge

C Scattering of α-particle by free electrons

D Scattering of α-particle by bound electrons

31. [VJC/H2/Prelims 2010/P1/40]

A radioactive source contains the nuclide $^{187}_{74}W$ which has a half life of 24 hours. In the absence of this source, a constant average count-rate of 10 min^{-1} is recorded.

Immediately after the source is placed in a fixed position near the counter, the average count-rate rises to 90 min^{-1}.

What average count-rate is expected with the source still in place 12 hours later?

A 45 min^{-1} B 50 min^{-1} C 57 min^{-1} D 67 min^{-1}

32. **[YJC/H2/Prelims 2010/P1/39]**

A Thorium-234 nucleus decays and undergoes two beta-particle emissions, one alpha-particle emission and one unknown emission to form the Radium-226 nucleus.

What could be the unknown emission?

A Alpha-particle emission

B Beta-particle emission

C Gamma emission

D Neutron emission

33. **[YJC/H2/Prelims 2010/P1/40]**

A stationary uranium nucleus, $^{238}_{92}U$ undergoes radioactive decay with emission of a helium nucleus, $^{4}_{2}He$ of kinetic energy *E*.

What is the kinetic energy of the daughter nucleus?

A $\frac{4}{234}E$ B $\frac{4}{238}E$ C E D $\frac{238}{4}E$

II. Structured Questions

1. **[CJC/H2/Prelims 2010/P3/3]**

When beryllium is bombarded with α-particles of energy 8.0 × 10^{-13} J, carbon atoms are produced, together with a very penetrating radiation. A student suggested that the nuclear reaction might be

$$^{9}_{4}Be + ^{4}_{2}He \rightarrow ^{13}_{6}C + \gamma .$$

a. Explain what is meant by $^{13}_{6}C$. [1]

b. **i.** With the following information, calculate the energy released in the reaction as suggested above. [2]

nuclide	mass / u
$^{9}_{4}Be$	9.0150
$^{4}_{2}He$	4.0040
$^{13}_{6}C$	13.0075

ii. The energy of the penetrating radiation is found to be at least 8.8 × 10^{-12} J. Explain why the student's suggestion cannot be valid. [1]

[Total: 4]

2. **[Dunman High/H2/Prelims 2010/P3/5]**

A stationary radon (Rn) nucleus may decay spontaneously into a polonium (Po) nucleus and an α-particle as shown below. It may be assumed that no γ-ray is emitted in the process.

$$^{222}_{86}Rn \rightarrow ^{218}_{84}Po + ^{4}_{2}He$$

The rest masses of the polonium nuclei and constituent nucleons are

^{218}Po: 218.0090 u

proton: 1.007276 u

neutron: 1.008664 u

a. Explain what is meant by *spontaneous*. [1]

b. Show that the binding energy per nucleon of Polonium is about 1.208 × 10^{-12} J. [3]

c. The binding energy per nucleon of ^{222}Rn is 1.201 × 10^{-12} J and that of ^{4}He is 1.092 × 10^{-12} J. Calculate the total kinetic energy of the decay. [2]

[Total: 6]

3. **[HCI/H2/Prelims 2010/P3/4]**

In 1932, Cockcroft and Walton produced nuclear disintegrations by bombarding lithium with the high speed protons. The protons were accelerated through a potential difference of 4.00×10^5 V using a specially built high-voltage machine. Photographs of the reaction taken in a cloud chamber show that two alpha particles were produced. The tracks were straight and as their range was equal, the alpha particles have the same initial energy. Using the length of the tracks, the initial energy of the alpha particles was calculated. The experimental value of the energy agreed closely to the theoretical value, providing the earliest verification of Einstein's mass-energy relation.

The nuclear reaction is given by

$$ {}^7_3\text{Li} + {}^1_1\text{H} \rightarrow 2\,{}^4_2\text{He}\,. $$

a. i. Explain what is meant by ${}^7_3\text{Li}$. [1]

ii. By considering the Coulomb repulsion between the lithium nucleus and the proton, calculate the distance of closest approach between the lithium nucleus and the accelerated proton. [2]

b. The masses of the nuclei involved are listed below:

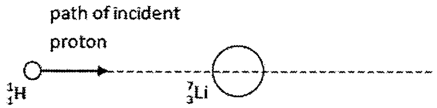
${}^7_3\text{Li}$	7.0138 u
${}^4_2\text{He}$	4.0015 u
${}^1_1\text{H}$	1.0073 u

i. Ignoring the kinetic energy of the proton, calculate the energy of each alpha particle. [3]

ii. On average, an alpha particle creates 5.0×10^3 ion pairs per mm of track in the cloud chamber and the energy needed to produce an ion pair is 5.2×10^{-18} J.

1. Calculate the length of the track made by the alpha particle. [1]

2. Sketch the tracks produced by the two alpha particles in figure below. [1]



c. In 1934, Fermi began using neutrons instead of protons to produce nuclear disintegrations. Neutrons are generally more effective than protons or alpha particles for this purpose. Suggest a reason why this may be so. [1]

[Total: 9]

4. **[IJC/H2/Prelims 2010/P3/6]**

a. i. Draw a labelled diagram of the type of apparatus used by Rutherford, Geiger and Marsden to investigate the nuclear model of the atom. [2]

ii. With the aid of diagrams, discuss qualitatively two important outcomes of the experiment that provided the evidence for the nuclear model of the atom. [6]

b. A stationary radium (${}^{226}_{88}\text{Ra}$) nucleus decays into a radon (Rn) nucleus and an α-particle.

i. Write an equation for the radioactive decay. [1]

ii. Show that the ratio of speeds $\frac{v_\alpha}{v_{Rn}}$ after the decay is approximately 56. [2]

iii. After the decay which takes place at a large distance from the gold nucleus, the α-particle moves head-on towards a gold (${}^{197}_{79}\text{Au}$) nucleus with a speed of 1.66×10^7 m s^{-1}.

Sketch 2 curves on Fig. 4.1 to illustrate how the electrical potential energy between the two particles, and the kinetic energy of the α-particle varies with position between the α-particle and the gold nucleus.

You may assume that the gold nucleus remains stationary during the interaction [2]

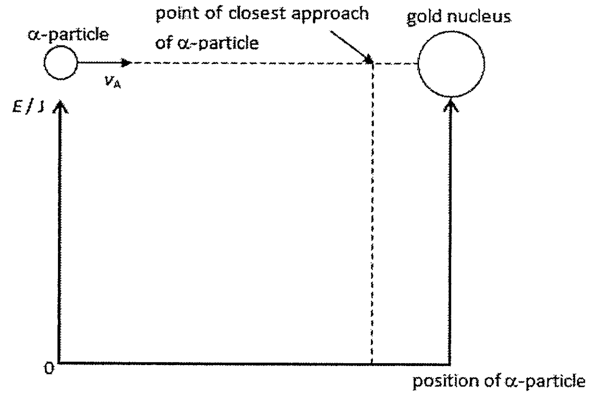


Fig. 4.1

iv. Calculate the distance of closest approach of the α-particle relative to the gold nucleus. [3]

v. With reference to your answer to **b.iv.**, comment on the radius of the gold nucleus. [1]

vi. Estimate the ratio of

1. the diameter of gold atom to the diameter of its nucleus [1]

2. the mass of a gold nucleus to the mass of a gold atom. [2]

[Total: 20]

5. **[MI/H2/Prelims 2010/P3/6 (part)]**

a. Fig. 5.1 shows the path of an α-particle as it passes near the nucleus of a gold atom.

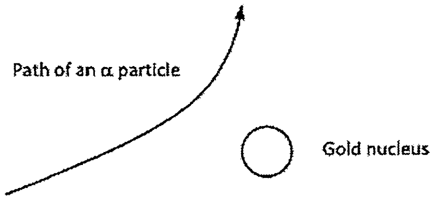


Fig. 5.1

i. Explain why the α-particle was deflected as shown in Fig. 5.1. [2]

ii. Indicate on the diagram the direction of the electric force acting on the α-particle. [1]

b. Xenon-139 has a half-life of 41 s and is generated at a constant rate during the fission of a specific sample of Uranium-235. The number of Xenon-139 nuclei in the sample increases initially and finally becomes constant.

i. Explain the meaning of the following terms:

1. half-life [1]

2. fission. [1]

ii. Suggest a reason why the number of Xenon-139 nuclei in the sample becomes constant. [1]

iii. The activity of Xenon-139 is 3.4×10^8 Bq when the number of Xenon-139 nuclei has reached a constant. Calculate

1. the number of Xenon-139 nuclei present in the sample [2]

2. the mass of Xenon-139 in the sample. [2]

c. The energy released in the fission reaction of Uranium-235 occurs partly as kinetic energy of the fission products (167 MeV) and of the neutrons (5 MeV).

In a nuclear power station, 25% of the energy of the *fission products* is converted into electrical energy. The number of uranium nuclei in 1.0 kg of Uranium-235 is 2.56×10^{24}.

i. Calculate the electrical energy generated from the fission of 1.0 kg of Uranium-235. [2]

ii. Calculate the average power output of the power station if the duration of the fission reaction of Uranium-235 is 24 hours. [2]

[Total: 14]

6. **[MJC/H2/Prelims 2010/P2/5 (modified)]**

a. Explain what is meant by *binding energy*. [1]

b. Calculate the binding energy of a thorium nucleus $^{226}_{90}Th$.

rest mass of $^{226}_{90}Th$ = 226.0249 u

rest mass of proton = 1.0073 u

rest mass of neutron = 1.0087 u [2]

c. A thorium nucleus $^{226}_{90}Th$ originally at rest decays and forms a radium nucleus $^{222}_{88}Ra^*$ and an alpha particle as shown below. The radium nucleus $^{222}_{88}Ra^*$ is in an excited state.

$$^{226}_{90}Th \rightarrow \, ^{222}_{88}Ra^* + \, ^{4}_{2}He$$

rest mass of $^{226}_{90}Th$ = 226.0249 u

rest mass of $^{222}_{88}Ra$ = 222.0154 u

rest mass of $^{4}_{2}He$ = 4.0026 u

Calculate the kinetic energy of the radium nucleus if the alpha particle is emitted with a kinetic energy of 6.31 MeV. [3]

[Total: 6]

7. **[NYJC/H2/Prelims 2010/P2/6]**

A stationary Polonium-212 nuclide may undergo alpha decay spontaneously to produce the stable lead-208 daughter nuclide as shown in the equation below:

$$^{212}_{84}Po \rightarrow \, ^{208}_{82}Pb + \, ^{4}_{2}He$$

The rest masses of these nuclei are $^{212}_{84}Po$: 211.9888 u; $^{208}_{82}Pb$: 207.9766 u and $^{4}_{2}He$: 4.0026 u.

a. Calculate the total kinetic energy of the decay products. [2]

b. Determine the ratio $\dfrac{\text{kinetic energy of He-4}}{\text{kinetic energy of Pb-208}}$. Give your answer to 3 significant figures. [2]

c. Hence, determine the kinetic energy of the alpha particle in MeV. [2]

[Total: 6]

8. [NJC/H2/Prelims 2010/P3/5]

a. $^{226}_{88}Ra$ is a stationary radioactive isotope which decays to $^{222}_{86}Rn$ with the release of an alpha particle.

(Mass of $^{4}_{2}He$ = 4.00260 u, mass of proton = 1.00783 u, mass of neutron = 1.00867 u, mass of $^{222}_{86}Rn$ = 222.018 u)

i. Define *binding energy of a nucleus* and explain how this quantity could be a measure of the stability of a nuclide. **[2]**

ii. The binding energy per nucleon of $^{226}_{88}Ra$ is 7.66831 MeV. Show that its mass is 226.03 u. **[2]**

iii. Starting from first principles, show that

$$Q = K_{\alpha}\left(1+\frac{m_{\alpha}}{m_{Rn}}\right),$$

where Q is the energy released in the decay reaction, K_{α} is the kinetic energy of the alpha particle, m_{α} is the mass of the alpha particle and m_{Rn} is the mass of $^{222}_{86}Rn$. **[3]**

iv. Hence, calculate K_{α}. **[3]**

v. Radon (Rn) decays by alpha emission to polonium and a tube containing an isotope of radon is to be implanted in a patient. Suggest and explain two reasons why an alpha emitter is preferred to the beta or gamma emitter for such purpose. **[2]**

b. Fig. 8.1 show the activity A of two samples of sodium nuclides, X and Y.

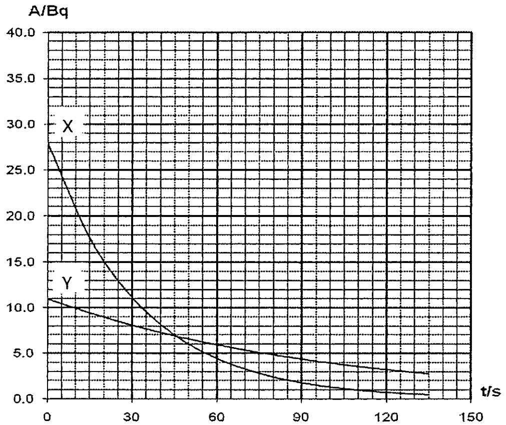


Fig. 8.1

i. Define *activity* and *half-life* of a radioactive nuclide. **[2]**

ii. Determine the ratio $\dfrac{\text{number of undecayed X nuclei}}{\text{number of undecayed Y nuclei}}$ when the activities of the two samples are the same. **[2]**

iii. How would you tell from the graphs, as drawn, that the background radiation is negligible? **[2]**

iv. Explain clearly how you would show that the activity of the nuclides decay exponentially. **[2]**

[Total: 20]

9. [PJC/H2/Prelims 2010/P2/5]

a. Explain what is meant by

i. *binding energy* of a nucleus **[2]**

ii. *nuclear fission.* **[2]**

b. The binding energy per nucleon varies with nucleon number in the way shown in Fig. 9.1.

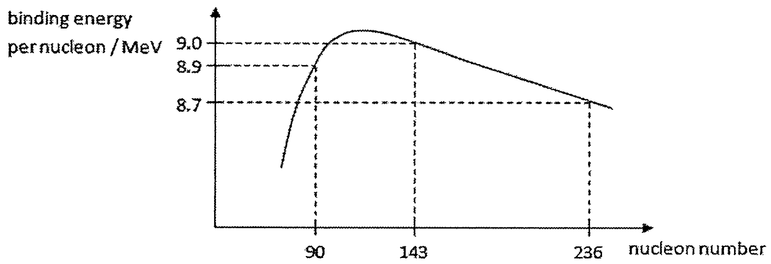


Fig. 9.1

During one particular fission process, a Uranium-236 nucleus gives, among its fission products, a Strontium-90 nucleus and a Xenon-143 nucleus. The equation is given by

$$^{236}_{92}U + ^{1}_{0}n \rightarrow ^{90}_{38}Sr + ^{143}_{54}Xe + 4\,^{1}_{0}n + \text{energy}.$$

i. Use the values on Fig. 9.1 to calculate the energy released during this fission process. **[2]**

ii. Explain the role of the neutrons in the fission reaction and why are these neutrons not taken into account in the calculation in **b.i.**. **[2]**

iii. Explain why a release of energy occur when there is an increase in the binding energy. **[4]**

[Total: 10]

10. [RVHS/H2/Prelims 2010/P2/6]

The isotope Iron-59 is a β-emitter with a half-life of 45 days. In order to estimate engine wear, an engine component is manufactured from non-radioactive iron throughout which the isotope Iron-59 has been uniformly distributed. The mass of the component is 2.4 kg and its initial activity is 8.5×10^{7} Bq.

The component is installed in the engine 60 days after manufacture of the component, and then the engine is tested for 30 days. During the testing period, any metal worn off the component is retained in the surrounding oil. Immediately after the test, the oil is found to have a total activity of 880 Bq.

Calculate

a. the decay constant for the isotope Iron-59 **[2]**

b. the total activity of the component when it was installed **[2]**

c. the mass of iron worn off the component during the test. **[3]**

[Total: 7]

11. [RI/H2/Prelims 2010/P3/7]

a. Define *binding energy*. [2]

b. A minimum energy Q is required to remove a neutron from a helium-4 nuclide to form a helium-3 nuclide. The following data is given:

Binding energy per nucleon of helium-4 nuclide = 6.8465 MeV

Binding energy per nucleon of helium-3 nuclide = 2.2666 MeV

Mass of neutron = 1.0087 u

1 u ≡ 931.494 MeV

i. Write the nuclear equation for this reaction. [1]

ii. Calculate Q. [3]

iii. Hence, calculate the difference in mass between the helium-3 and helium-4 nuclides. [3]

iv. With reference to the above process, explain why the mass difference is less than the mass of a neutron. [1]

c. Helium-2 is a hypothetical isotope of helium which consists of two protons and no neutrons. It has *negative binding energy*.

i. Explain the implication of the italicized terms on helium-2. [1]

ii. Suggest a reason for this. [1]

d. A radioactive source contains a mixture of nuclides ${}^{32}_{15}P$ and ${}^{35}_{16}S$. The half-life of ${}^{35}_{16}S$ is 6.14 times that of ${}^{32}_{15}P$. At time t = 0 s, 90% of the total activity from this source comes from the ${}^{32}_{15}P$ nuclides.

i. Define *half-life*. [1]

ii. If the number of ${}^{32}_{15}P$ nuclides is N at time t = 0 s, determine the number of ${}^{35}_{16}S$ nuclides in the source in terms of N. [3]

iii. Calculate the time elapsed for 90% of the total activity to come from the ${}^{35}_{16}S$ nuclides, given that the half-life of ${}^{32}_{15}P$ is 14.3 days. [4]

iv. Suggest a possible use for ${}^{32}_{15}P$, which is a beta-emitter with a half-life of 14.3 days. [1]

[Total: 20]

12. [SAJC/H2/Prelims 2010/P3/1]

a. For a spacecraft launched into the outer regions of the solar system, it is not practical to have its battery recharged by solar panels. Such spacecrafts use Plutonium-238 (Pu-238), which is an alpha emitter with a half-life of 88 years, as fuel.

i. If each alpha particle is emitted with a kinetic energy of 5.0 MeV, calculate that the minimum activity of the source required to produce an alpha particle beam of 20 W. [2]

ii. Show that the decay constant, λ, of Pu-238 is 2.5×10^{-10} s^{-1}. [1]

iii. Calculate the mass of Pu-238 required to generate the activity shown in **i.**. [3]

iv. Plutonium is one of the most dangerous radioactive substances known. It has been estimated that even a small amount of this substance, suitably distributed, would be enough to kill more than a billion people. Comment on the risks on the population on Earth involved in using plutonium as a fuel for spacecraft. [2]

b. The radioactive isotope, Iodine-131, with a half-life of 30 years, and Caesium-137, with a half-life of 8 days, are the nuclear wastes generated from nuclear power plants.

Consider that each of these wastes is of similar mass and sufficient to pose a hazard initially, explain which of these nuclear wastes should be stored for a longer period of time. [2]

[Total: 10]

13. [TJC/H2/Prelims 2010/P2/6]

The following extract is taken from a science fiction magazine:

> It is the distant future and Man has long since abandoned Planet Earth. Space explorers land on the Earth's surface and discover a nuclear warhead manufactured in the 20th century. They analyse a sample of the warhead material and find that it contains a mixture of radioactive plutonium Pu-239 and effectively stable uranium U-235.
>
> The results obtained for the sample are:
>
> mass of ${}^{239}_{94}Pu$ present = 2.0×10^{-6} kg
>
> mass of ${}^{235}_{92}U$ present = 6.0×10^{-6} kg
>
> activity of ${}^{239}_{94}Pu$ in sample = 4.4×10^{6} disintegrations per second

a. State what is meant by *activity* of a radioactive sample. [1]

b. Show that the number of ${}^{239}_{94}Pu$ atoms in the sample is 5.0×10^{18}. [1]

c. Calculate the radioactive decay constant for ${}^{239}_{94}Pu$. [2]

d. Explorers assume that the material was originally pure plutonium which decayed to produce the uranium.

i. Show that the fraction of the original plutonium atoms that remains undecayed is 0.25. [2]

ii. Hence calculate, in years, how far into the future the extract is set. [3]

[Total: 9]

14. [TJC/H2/Prelims 2010/P3/4]

a. **i.** Explain what is meant by *nuclear binding energy.* **[1]**

ii. On the axes in Fig. 14.1, sketch the variation of the binding energy per nucleon with nucleon number. **[1]**

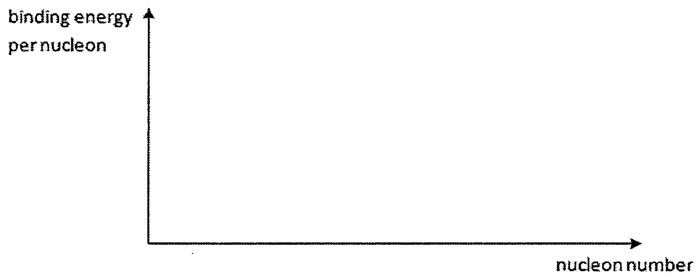


Fig. 14.1

iii. By reference to your graph drawn in **a.ii.**, explain how fission can be a potential source of energy. **[2]**

b. When a uranium-235 nucleus undergoes fission, two nuclides are produced with the release of energy as shown in the equation:

$$^{235}_{92}U+{}^{1}_{0}n\rightarrow {}^{139}_{54}Xe+{}^{95}_{38}Sr+2{}^{1}_{0}n.$$

The masses of the nuclides are as follows:

nuclide	mass / u
$^{235}_{92}U$	235.043929
$^{139}_{54}Xe$	138.918793
$^{95}_{38}Sr$	94.919359
$^{1}_{0}n$	1.008665

i. Determine the energy released in one reaction. **[3]**

ii. Singapore's yearly energy consumption in 2007 was approximately 1.15×10^{17} J.

Assuming an efficiency of 8.0%, determine how long, in days, the energy released from the fission of 2000 kg of uranium can be used to power the city of Singapore. **[3]**

[Total: 10]

15. [VJC/H2/Prelims 2010/P3/6]

a. **i.** Define the term *binding energy.* **[1]**

ii. Use the data below to calculate the binding energy in MeV of a nucleus of $^{16}_{8}O$.

Data: mass of proton = 1.007 276 u

mass of neutron = 1.008 665 u

mass of oxygen nucleus = 15.990 527 u **[3]**

iii. The binding energy of $^{17}_{8}O$ is 126.43 MeV.

State which of these two isotopes of oxygen would be more stable. Explain your answer. **[4]**

b. Scientists have worked out the age of the Moon by dating rocks brought back by the Apollo missions. They use the decay of potassium $^{40}_{19}K$ to argon $^{40}_{18}Ar$, which is stable. The decay constant of potassium-40 is 5.3×10^{-10} per year.

i. Write the full nuclear equation for this decay. **[2]**

ii. Explain what is meant by the decay constant of potassium-40 being 5.3×10^{-10} per year. **[1]**

iii. Sketch a labelled graph to show how the activity of $^{40}_{19}K$ changes with time. Mark on your graph the half-life of $^{40}_{19}K$. **[2]**

iv. In one rock sample the scientists found 0.84 μg of argon-40 and 0.10 μg of potassium-40. Calculate the age of the rock sample in years. **[3]**

v. Stat any assumption that you have made for the calculation in **iv.**. **[1]**

vi. Calculate the activity of the potassium-40 in the rock sample. Hence, explain if it is necessary for the scientists who handled the rock sample to take special safety precautions. **[3]**

[Total: 20]

16. [YJC/H2/Prelims 2010/P3/8]

a. Explain what is meant by *nuclear fission.* **[2]**

b. A typical nuclear fission reaction that involves uranium-235 is represented by the equation

$$^{235}_{92}U+n\rightarrow {}^{92}_{36}Kr+{}^{141}_{56}Ba+3n.$$

nuclide	mass / u
$^{235}_{92}U$	235.044
$^{92}_{36}Kr$	91.910
$^{141}_{56}Ba$	140.916
n	1.009

i. Deduce the number of protons and neutrons in the $^{92}_{36}Kr$ nucleus. **[2]**

ii. Calculate the energy released, in joules, in the above reaction. **[3]**

iii. If a nuclear power station uses up $^{235}_{92}U$ at a rate of 3.5×10^{-3} kg s^{-1} and has an efficiency of 23%, estimate the useful power output. **[3]**

iv. State two forms of energy of the product particles. **[2]**

c. Uranium-234 is another isotope of uranium that is radioactive and has a half-life of 2.4×10^{5} years. The daughter nuclei from the decay is Thorium-230 with the emission of another particle **X**.

i. Define half-life. **[1]**

ii. State what **X** is. **[1]**

iii. Calculate the decay constant of Uranium-234. **[1]**

iv. Calculate the activity of a U-234 source after 8.7×10^{4} years if it initially has 5.5×10^{26} atoms. **[3]**

v. Describe two applications of radioisotopes. **[2]**

[Total: 20]

	Raffles Institution Physics Department Revision Package 1	**Nuclear Physics**	**20**

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	D	A	C	C	D	A	B	B	A	C	B	C	D	D

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	D	B	C	C	D	D	C	D	A	A	B	A	B	B

31	32	33
D	A	A

1 C Alpha particle is only deflected significantly when it passes by very close (small impact parameter) to the nucleus. Since most of the alpha particles pass through undeflected (or with very little deflection), it implies that although these alpha particles pass through the empty spaces within the atom and are nowhere close to the extremely small nucleus.

2 D During a nuclear fission, a heavy nucleus (parent) breaks up into two or more lighter daughter nucleus which are energetically more stable (higher binding energies) than the parent. Some neutrons (zero binding energy) may also be emitted.

3 A During the Rutherford α-scattering experiment, most of the alpha particles are undeflected or deflected very slightly. Only a handful of alpha particles are deflected at large angles ($>90°$).

4 C Using $A=\lambda N$ and $t_{1/2}=\ln 2/\lambda$, $t_{1/2}=\dfrac{N\ln 2}{A}=\dfrac{9.49\times10^{19}\ln 2}{1600}=4.11\times10^{16}$ s

5 C By comparing the mass number (top) and atomic number (bottom) on LHS and RHS, X has a mass number of 1 and atomic number of 0. So X is a neutron.

6 D If sodium and aluminium are fused together (with enough energy), the product is a more stable heavier nucleus (higher binding energy) and net energy will be released.

7 A Since the energy of an electron and a positron is converted into two identical photons, each photon has a minimum energy of mc^2 (assuming both the electron and positron are at rest; greater if they are moving).
So the wavelength of the photons is

$$\lambda=\frac{hc}{E}\geq\frac{hc}{mc^2}=\frac{h}{mc}$$

8 B Sample's initial activity is $A_0=\lambda N=\dfrac{N\ln 2}{t_{1/2}}=\dfrac{10^{18}\ln 2}{2.00\times24\times60\times60}=4.01\times10^{12}\ \text{s}^{-1}$

After 5 days, sample activity is $A=A_0\left(\frac{1}{2}\right)^{5/2}=4.01\times10^{12}\left(\frac{1}{2}\right)^{5/2}=7.09\times10^{11}\ \text{s}^{-1}$

9 B The sheet of paper blocks alpha particles. So it can be concluded that the initial count rate of material A (emitting alpha particles) is 64 Bq while that of material B (emitting beta particles) is 96 Bq.
After 12 days, count rate of A drops to 1/8 of its initial, i.e., 8 Bq (3 half-lives) while that of B drops to 1/16 of its initial, i.e., 6 Bq (4 half-lives). So the total count rate after 12 days is 14 Bq.

10 A A slow neutron has zero kinetic energy and also zero binding energy. So the energy of the γ-ray is given by

$$E=\text{BE}_{\text{Li}}+\text{BE}_{\text{He}}-\text{KE}_{\text{Li}}--\text{KE}_{\text{He}}-\text{BE}_{\text{B}}=39.25+28.48-2.31-64.94=0.48\ \text{MeV}$$

Explanation: Binding energy is defined as the energy needed to break up a nucleus into its constituent particles (protons and neutrons). So 64.94 MeV is needed to break up each Boron nucleus. However, these particles then form Lithium and Helium nucleus which releasing energies numerically equal to their respectively binding energy. So the excess energy becomes kinetic energy of the products and γ-ray.

11 C Using $A=A_0\left(\frac{1}{2}\right)^n$ where n is the number of half-lives, we obtain

$$n=\frac{\ln(A/A_0)}{\ln\left(\frac{1}{2}\right)}=\frac{\ln(0.1)}{\ln\left(\frac{1}{2}\right)}=3.32$$

So the time elapsed is 9.96 s.

12 B Only equations A and B show alpha particles causing a nuclear reaction but ${}^{1}_{1}\text{n}$ is incorrect.

13 C After 3 half-lives, the number of nuclei remaining is $(1/2)^3=1/8$ of its initial number of nuclei. So the number of nuclei decayed is $7N_0/8$.

14 D The activity of the injected iodine-123 after 24 hours is $A=A_0\left(\frac{1}{2}\right)^{24/13}=8.34\ \mu\text{Ci}$

So the fraction of iodine concentrated in thyroid is $f=\dfrac{4.00}{8.34}=0.48$

15 D Almost all the alpha particles are undeflected or deflected very little. A very small number are deflected at large angles.

16 D The initial count rate due to the source is $68\ s^{-1}$ (adding the background count rate of $10\ s^{-1}$).

After 40 minutes (2 half-lives), the count rate is $17\ s^{-1}$ at a distance of 80 cm from source. If the detector is brought to a distance of 40 cm, the count rate due to the source would be 4x since count rate $\propto 1/r^2$ (as particles are spread over a spherical surface area of $4\pi r^2$).

So total count rate is $78\ s^{-1}$ (adding the background count rate of $10\ s^{-1}$).

17 D Number of daughter nucleus increases with decreasing rate: $N = N_0\left(1-e^{-\lambda t}\right)$.

Rate of alpha emission is proportional to number of parent nuclei present: $A = \lambda N$.

18 B A nucleus with a greater mass defect is more stable. If the products have a greater total mass defect than the original nucleus, then energy is released during the nuclear reaction.

19 C The count rate due to 1 g of living wood and specimen are $55\ \text{min}^{-1}$ and $10\ \text{min}^{-1}$, respectively. So the number of half-lives that has elapsed is

$$n = \frac{\ln\left(C_{\text{specimen}}/C_{\text{living}}\right)}{\ln\left(\frac{1}{2}\right)} = \frac{\ln(10/55)}{\ln\left(\frac{1}{2}\right)} = 2.46$$

So the approximate age of specimen is $2.46 \times 5700 = 14000$ y

20 C Fraction of sample remaining $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{16/32} = 0.707$

21 D The nucleus equation is ${}^{14}_{7}\text{N} + {}^{4}_{2}\text{He} \rightarrow {}^{17}_{8}\text{O} + {}^{1}_{1}\text{p}$.

Note that γ-photon is usually emitted as the daughter nucleus may be in an excited state.

22 D Let initial activity of X be A. After 6 half-lives of X, activity of X is $A \times 1/2^6 = A/64$.

Let initial activity of Y be $2A$.

In the same time, 3 half-lives of Y have elapsed. So activity of Y is $2A \times 1/2^3 = A/4$.

So ratio of activity of X to Y is 1:16.

23 C For energy release, the total binding energy of the product must be greater than that of the reactants.

24 D Total binding energy of ${}^{156}_{64}\text{Gd}$ is $\text{BE} = 156 \times 1.3 = 203$ pJ

25 A After 3 half-lives, the number of nuclei remaining is $(1/2)^3 = 1/8$ of its initial number of nuclei. So the number of nuclei decayed is $7N_0/8$.

26 A Energy released $= (143+90)x - 235 \times 7.6 = 195$ MeV

$$x = \frac{195 + 235 \times 7.6}{143+90} = 8.5\ \text{MeV}$$

27 B After 16 years, $\frac{N}{N_0} = \frac{31250}{500000} = \frac{1}{16}$

This means that 4 half-lives have elapsed. Therefore half-life of cobalt is 4 years.

After 12 years, the number of undecayed Cobalt nuclei is $500000/2^3 = 62500$.

28 A Mass is converted into energy in this nuclear reaction. The mass converted to energy is

$$m = \frac{E}{c^2} = \frac{3.27 \times 1.60 \times 10^{-13}}{\left(3.00 \times 10^8\right)^2} = 5.81 \times 10^{-30}\ \text{kg}$$

So the mass of two deuterium nuclei is 5.81×10^{-30} kg greater than the combined mass of products.

29 B Let the initial activity of X and Y be $2.5A$ and A, respectively.

After 3 days, activity of Y is A/4 $\Rightarrow$ activity of X is 1.25A.

Therefore half-life of X is 3 days.

30 B This is the Rutherford α-scattering experiment.

31 D The count rate due to the source is $80\ \text{min}^{-1}$

After 12 hours, the count rate due to the source is $A = A_0\left(\frac{1}{2}\right)^{12/24} = 80 \times 0.707 = 57\ \text{min}^{-1}$

Therefore the total count rate is $67\ \text{min}^{-1}$.

32 A The mass number decreases by 8. So the unknown emission causes the mass number to decrease by 4 (alpha emission) since beta emission does not affect the mass number of the nucleus.

33 A By the principle of conservation of momentum,

$$m_d v_d = m_\alpha v_\alpha \quad \Rightarrow \quad \frac{v_d}{v_\alpha} = \frac{m_\alpha}{m_d}$$

$$\frac{K_d}{K_\alpha} = \frac{\frac{1}{2} m_d v_d^2}{\frac{1}{2} m_\alpha v_\alpha^2} = \frac{\frac{1}{2} m_d}{\frac{1}{2} m_\alpha} \times \left(\frac{m_\alpha}{m_d}\right)^2 = \frac{m_\alpha}{m_d} = \frac{4}{234}$$

Structured Questions

1. **[CJC/H2/Prelims 2010/P3/3]**

a. $^{13}_{6}C$ denotes a particular carbon nuclide in which there are 13 nucleons, of which 6 are protons. Thus, there are (13 – 6) = 7 neutrons in that nuclide.

b. i. decrease in mass in the reaction: $\Delta m = (9.0150\text{ u} + 4.0040\text{ u}) - 13.0075\text{ u} = 0.0115\text{ u}$

energy released in the reaction: $E = \Delta mc^2 = 0.0115 \times (1.66\times10^{-27}) \times (3.00\times10^8)^2 = 1.72\times10^{-12}$ J

ii. The suggested reaction only releases 1.72×10^{-12} J of energy which is less than the observed value of 8.8×10^{-12} J. Hence, the suggested reaction does not agree with experimental results and so is invalid.

2. **[Dunman High/H2/Prelims 2010/P3/5]**

a. See definitions in lecture notes.

b. mass defect $\Delta m = 84 \times (1.007276) + (218 - 84) \times (1.008664) - 218.0090 = 1.7632$ u

binding energy $\text{B.E.} = \Delta mc^2 = (1.76316 \times 1.66\times10^{-27}) \times (3.00\times10^8)^2 = 2.63416\times10^{-10}$ J

binding energy per nucleon $= \dfrac{2.63416\times10^{-10}}{218} = 1.20833\times10^{-12} = 1.208\times10^{-12}$ J

c. $\Delta E = 2.63416\times10^{-10} + 4\times(1.092\times10^{-12}) - 222\times(1.201\times10^{-12}) = 1.162\times10^{-12}$ J

Since $\Delta E > 0$, energy is released from the reaction. This release of energy manifests as the total kinetic energy of the particles after the reaction. Thus, total kinetic energy = 1.162×10^{-12} J

3. a. i. A nuclide with 3 protons and 4 neutrons.

ii. Kinetic energy of the proton after being accelerated through the potential difference:

$$E_k = q\Delta V = e\Delta V$$

Let x be the distance of closest approach. By conservation of energy,

$$\Delta E_p = -\Delta E_k$$

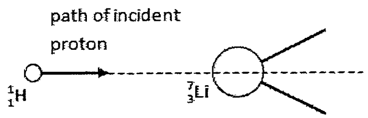
$$\frac{Qe}{4\pi\varepsilon_0 x} - 0 = -(0 - e\Delta V) \quad \text{where } Q \text{ is the charge of the lithium nucleus}$$

$$x = \frac{Q}{4\pi\varepsilon_0\Delta V} = \frac{3\times1.60\times10^{-19}}{4\pi\varepsilon_0\times400000} = 1.08\times10^{-14}\text{ m}$$

b. i. $E = \Delta mc^2 = [7.01381.0073 - (2\times4.0015)]\times1.66\times10^{-27}\times(3.00\times10^8)^2 = 2.70\times10^{-12}$ J

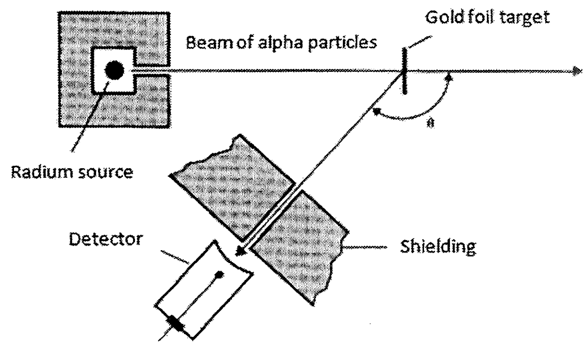
energy of each alpha particle = $\dfrac{2.70414\times10^{-12}}{2} = 1.35\times10^{-12}$ J

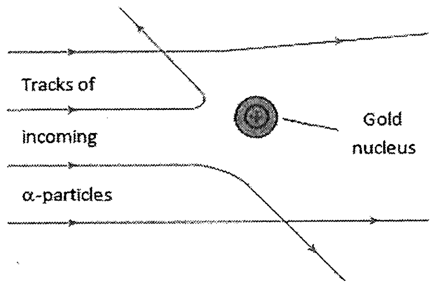
ii. 1. length= $\dfrac{1.35207\times10^{-12}}{(5.0\times10^3)(5.2\times10^{-18})} = 52.0$ mm

2. 

c. Neutrons have no charge and are therefore able to penetrate more deeply into the positively charged nucleus, resulting in higher probability of nuclear reactions.

4. **[IJC/H2/Prelims 2010/P3/6]**

a. i. 

ii. 

Most of the α-particles pass through the foil with little deflection and a very small proportion of the α-particles, about 1 in 8000, are deflected at large angles. This indicates that all the positive charge of the atom and most of the mass was concentrated in a tiny space inside the atom called the nucleus and that the gold nucleus is not just extremely small but also very massive.

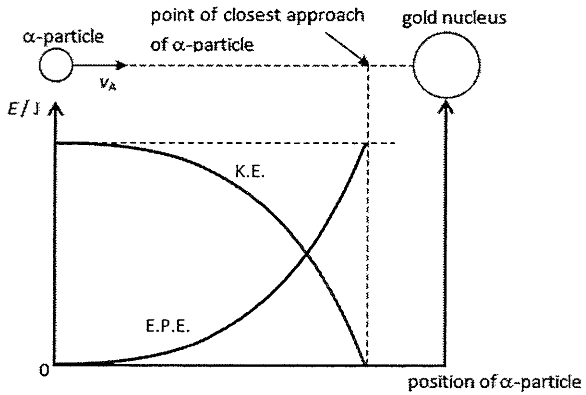
b. i. $^{226}_{88}Ra \rightarrow ^{222}_{86}Rn + ^4_2He$

ii. By conservation of momentum,

$$0 = m_\alpha v_\alpha + m_{Rn}(-v_{Rn})$$

$$\frac{v_\alpha}{v_{Rn}} = \frac{m_{Rn}}{m_\alpha} = \frac{222\text{ u}}{4\text{ u}} = 55.5 = 56$$

iii.



iv. By conservation of energy,

$$\Delta E_p = -\Delta E_k$$

$$\frac{q_\alpha q_{Au}}{4\pi\varepsilon_0 r} - 0 = -\left(0 - \frac{1}{2}m_\alpha v_\alpha^2\right) \quad \text{where } r \text{ is the distance of closest approach}$$

$$r = \frac{(2e)(79e)}{2\pi\varepsilon_0 m_\alpha v_\alpha^2} = \frac{2\times79\times\left(1.60\times10^{-19}\right)^2}{2\pi\varepsilon_0\times\left(4\times1.66\times10^{-27}\right)\times\left(1.66\times10^{7}\right)^2} = 3.97\times10^{-14}\text{ m}$$

v. The radius of the gold nucleus is in the order of 10^{-14} m.

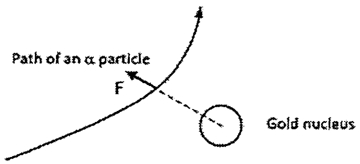
vi. 1. $\dfrac{d_{atom}}{d_{nucleus}} = \dfrac{10^{-10}}{10^{-14}} = 10^4$

2. $\dfrac{m_{nucleus}}{m_{atom}} = \dfrac{79m_{proton}+118m_{neutron}}{79m_{proton}+118m_{neutron}+79m_{electron}} = \dfrac{197\times\left(1.67\times10^{-27}\right)}{197\times\left(1.67\times10^{-27}\right)+79\times\left(9.11\times10^{-31}\right)} = 1.00$

5. **[MI/H2/Prelims 2010/P3/6 (part)]**

a. i. Both the α-particle and nucleus are positively charged.
The deflection is due to the Coulomb repulsion between like charges.

ii. Force vector from centre of gold nucleus to path



b. i. 1. See definitions in lecture notes.

2. See definitions in lecture notes.

ii. When the rate of production of Xenon-139 nuclei in the sample from the Uranium-235 fission process is equal to its rate of decay.

iii. 1. $A = \lambda N \Rightarrow N = \dfrac{A}{\lambda} = \dfrac{3.4\times10^8}{\frac{\ln 2}{41}} = 2.01\times10^{10}$

2 $m = \dfrac{N}{N_A}\times M_{molar} = \dfrac{2.0111\times10^{10}}{6.02\times10^{23}}\times0.139 = 4.64\times10^{-15}$ kg

c. i. $E = \dfrac{25}{100}\times2.56\times10^{24}\times167 = 1.0688\times10^{26}\text{ MeV} = 1.71\times10^{13}$ J

ii. $P = \dfrac{E}{t} = \dfrac{1.71008\times10^{13}}{24\times3600} = 1.98\times10^{8}$ W

6. **[MJC/H2/Prelims 2010/P2/5 (modified)]**

a. See definitions in lecture notes.

b. Mass defect $\Delta m = 90\times(1.0073)+(226-90)\times(1.0087)-(226.0249) = 1.8153$ u

binding energy $\text{B.E.} = \Delta mc^2 = \left(1.8153\times1.66\times10^{-27}\right)\times\left(3.00\times10^8\right)^2 = 2.71\times10^{-10}$ J

c. By conservation of energy,

$$m_{Th}c^2 = m_{Ra^*}c^2 + E_{k,Ra^*} + m_\alpha c^2 + E_{k,\alpha}$$

$$E_{k,Ra^*} = \left(m_{Th} - m_{Ra^*} - m_\alpha\right)c^2 - E_{k,\alpha}$$

$$= \left(226.0249\text{ u} - 222.0154\text{ u} - 4.0026\text{ u}\right)c^2 - \left(6.31\times10^6\times1.60\times10^{-19}\right) = 2.13\times10^{-14}\text{ J}$$

7. **[NYJC/H2/Prelims 2010/P2/6]**

a. change in mass during the reaction: $\Delta m = 207.9766 + 4.0026 - 211.9888 = -0.0096$ u
by conservation of mass-energy, energy is released during the reaction.
kinetic energy: $\Delta E = \Delta mc^2 = \left(0.0096\times1.66\times10^{-27}\right)\times\left(3.00\times10^8\right)^2 = 1.43\times10^{-12}$ J

b. By conservation of momentum, $0 = p_{Pb} - p_{He} \Rightarrow p_{Pb} = p_{He}$

$$\frac{\text{kinetic energy of He-4}}{\text{kinetic energy of Pb-208}} = \frac{\frac{p_{He}^2}{2m_{He}}}{\frac{p_{Pb}^2}{2m_{Pb}}} = \frac{m_{Pb}}{m_{He}} = \frac{207.9766}{4.0026} = 51.960$$

c. $\dfrac{E_{k,He}}{E_{k,Pb}} = \dfrac{E_{k,He}}{E_{k,T} - E_{k,He}} = 51.960$

$E_{k,He} = 51.960\left(E_{k,T} - E_{k,He}\right)$

$E_{k,He} = \dfrac{51.960E_{k,T}}{1+51.960} = \dfrac{51.960}{1+51.960}\times\left(1.4342\times10^{-12}\right) = 1.4071\times10^{-12}\text{ J} = 8.79$ Mev

8. [NJC/H2/Prelims 2010/P3/5]

a. i. See definitions in lecture notes.

The higher the binding energy per nucleon in a nucleus, the more stable the nucleus is.

ii. $\text{B.E.} = \Delta mc^2$

$226 \times 7.66831 \times 1.60 \times 10^{-13} = \left[(88 \times 1.00783) + (138 \times 1.00867) - M\right](1.66 \times 10^{-27})c^2$

$M = 226.0295\ \text{u} = 226.03\ \text{u}$

iii. By conservation of momentum, $0 = p_\alpha - p_{Rn} \Rightarrow p_\alpha = p_{Rn}$

By conservation of energy,

$$Q = K_\alpha + K_{Rn} = \frac{p_\alpha}{2m_\alpha} + \frac{p_{Rn}}{2m_{Rn}} = \frac{p_\alpha}{2m_\alpha} + \frac{p_\alpha}{2m_{Rn}} = \frac{p_\alpha}{2m_\alpha} + \frac{p_\alpha}{2m_\alpha}\frac{m_\alpha}{m_{Rn}} = \frac{p_\alpha}{2m_\alpha}\left(1 + \frac{m_\alpha}{m_{Rn}}\right) = K_\alpha\left(1 + \frac{m_\alpha}{m_{Rn}}\right)$$

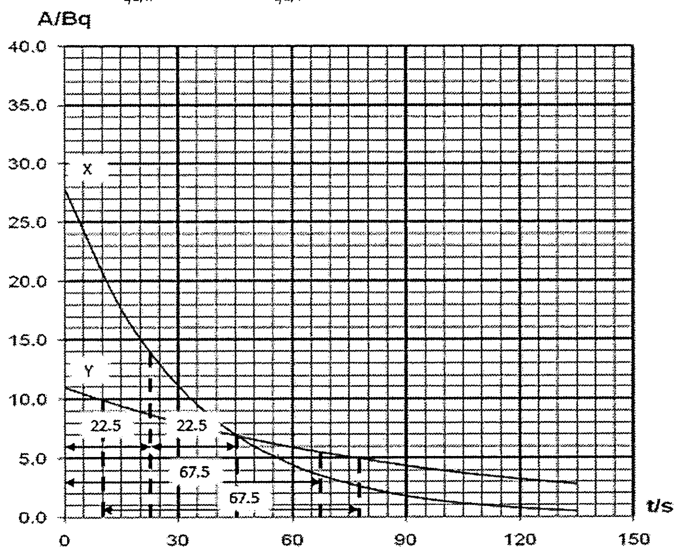
iv. $Q = \Delta mc^2 = (226.0295 - 222.018 + 4.00260) \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 1.3297 \times 10^{-12}\ \text{J}$

From **a.iii.**, $K_\alpha = \dfrac{Q}{1 + \frac{m_\alpha}{m_{Rn}}} = \dfrac{1.3297 \times 10^{-12}}{1 + \frac{4.00260}{222.018}} = 1.3062 \times 10^{-12}\ \text{J}$

v. Alpha radiation has low penetrating power and short ranges and is thus able to be used on localized area. It also has the highest ionization ability and is thus effective in killing harmful cells.

b. i. See definitions in lecture notes.

ii. From graph, we deduce that $t_{1/2,X} = 22.5\ \text{s}$ and $t_{1/2,Y} = 67.5\ \text{s}$.



$$A = \lambda_X N_X = \lambda_Y N_Y \quad \Rightarrow \quad \frac{N_X}{N_Y} = \frac{\lambda_Y}{\lambda_X} = \frac{\frac{\ln 2}{t_{1/2,Y}}}{\frac{\ln 2}{t_{1/2,X}}} = \frac{t_{1/2,X}}{t_{1/2,Y}} = \frac{22.5}{67.5} = 0.333$$

iii. activity becomes half after each half life.

iv. EITHER Show that 3 half-lives are constant

OR Explain that when ln(A/Bq) is plotted against t/s, the graph is a straight line.

9. [PJC/H2/Prelims 2010/P2/5]

a. i. See definitions in lecture notes.

ii. See definitions in lecture notes.

b. i. $\text{energy released} = \left[90 \times (8.9) + 143 \times (9.0)\right] - 236 \times (8.7) = 34.8\ \text{MeV}$

ii. The role of neutrons is to initiate more nuclear fission reactions by bombarding other $^{236}_{92}U$ nuclei such that a chain reaction results.

Neutrons have no binding energy.

iii. An increase in binding energy means that the product nuclei have higher binding energy per nucleon compared to the reactant nuclei. The products are more stable than the reactant.

The energy released when forming the products is more than the energy required when breaking the reactants into constituent particles.

10. [RVHS/H2/Prelims 2010/P2/6]

a. $\lambda = \dfrac{\ln 2}{t_{1/2}} = \dfrac{\ln 2}{45} = 0.0154\ \text{days}^{-1}$

b. $A_{60} = A_0 e^{-\lambda t} = (8.5 \times 10^7) \times e^{-0.01540 \times 60} = 3.37 \times 10^7\ \text{Bq}$

c. $A_{90} = A_0 e^{-\lambda t} = (8.5 \times 10^7) \times e^{-0.01540 \times 90} = 2.1256 \times 10^6\ \text{Bq}$

$\text{mass of iron worn off} = \dfrac{880}{2.1256 \times 10^6} \times 2.4 = 9.94 \times 10^{-4}\ \text{kg}$

11. [RI/H2/Prelims 2010/P3/7]

a. See definitions in lecture notes.

b. i. $^4_2\text{He} \rightarrow {}^3_2\text{He} + {}^1_0\text{n}$

ii. $E = (4 \times 6.8465) - (3 \times 2.2666) = 20.5862\ \text{MeV}$

iii. $E = \dfrac{20.5862}{931.494} = 0.0221002\ \text{u}$

$m_{^4_2He} - m_{^3_2He} = m_{^1_0n} - E = 1.0087 - 0.0221002 = 0.9866\ \text{u}$

iv. Energy is supplied in order to conserve mass-energy.

c. i. Helium-2 is unstable and cannot exist in a bound state.

ii. There is large Coulombic repulsion between the protons without any neutrons present to compensate by strong nuclear interactions.

d. i. See definitions in lecture notes.

ii. At time $t = 0$ s,

$$A_P = \lambda_P N \quad (1)$$

$$A_S = \lambda_S N_S \quad (2)$$

$$(1)\div(2): \quad \frac{A_P}{A_S} = \frac{\lambda_P N}{\lambda_S N_S}$$

$$\frac{90\%}{10\%} = \frac{\frac{\ln 2}{t_{1/2,\,P}}N}{\frac{\ln 2}{t_{1/2,\,S}}N_S}$$

$$9.0 = \frac{N}{t_{1/2,\,P}} \times \frac{6.14 t_{1/2,\,P}}{N_S}$$

$$N_S = \frac{6.14}{9.0}N = 0.682N$$

iii. After time t, activity of the $^{35}_{16}$S nuclides: $A_S = A_{S,0}e^{-\lambda_S t}$

After time t, activity of the $^{32}_{15}$P nuclides: $A_P = A_{P,0}e^{-\lambda_P t}$

$$\frac{A_S}{A_P} = \frac{A_{S,0}e^{-\lambda_S t}}{A_{P,0}e^{-\lambda_P t}} = \frac{A_{S,0}}{A_{P,0}} \times e^{(\lambda_P - \lambda_S)t}$$

$$\frac{90\%}{10\%} = \frac{10\%}{90\%} \times e^{(\lambda_P - \lambda_S)t}$$

$$(\lambda_P - \lambda_S)t = \ln(9.0^2)$$

$$t = \frac{2\ln 9.0}{\lambda_P - \lambda_S} = \frac{2\ln 9.0}{\ln 2 \times \left(\frac{1}{14.3 \text{ days}} - \frac{1}{6.14 \times 14.3 \text{ days}}\right)} = 108 \text{ days}$$

iv. It can be used to trace a plant's fertilizer uptake / as medical imaging and therapy / as tracers for body functions / to check thickness of materials.

12. [SAJC/H2/Prelims 2010/P3/1]

a. i. Energy carried by each alpha particle $= 5.0 \times 10^6 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-13}$ J

Activity required $= \dfrac{20}{8.0 \times 10^{-13}} = 2.5 \times 10^{13}$ s^{-1}

ii. $$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{88 \times 365 \times 24 \times 3600} = 2.50 \times 10^{-10} \text{ s}^{-1}$$

iii. $$N = \frac{A}{\lambda} = \frac{2.5 \times 10^{13}}{2.49768 \times 10^{-10}} = 1.00093 \times 10^{23}$$

$$m = \frac{N}{N_A} \times M = \frac{1.00093 \times 10^{23}}{6.02 \times 10^{23}} \times 0.238 = 0.0396 \text{ kg}$$

iv. Rockets may burn up in the atmosphere during launch or re-entry and failure. [1]

But plutonium would still be radioactive and when vaporised and dispersed widely in the atmosphere, it could be ingested.

b. Iodine-131 should be stored for a longer period of time because its activity decreases more slowly and hence will remain hazardous for a longer period of time.

13. [TJC/H2/Prelims 2010/P2/6]

a. See definitions in lecture notes.

b. $$N_{Pu} = \frac{m_{Pu}}{M_{Pu}} \times N_A = \frac{2.0 \times 10^{-6}}{0.239} \times 6.02 \times 10^{23} = 5.0377 \times 10^{18} = 5.0 \times 10^{18}$$

c. $$A_{Pu} = \lambda_{Pu} N_{Pu} \Rightarrow \lambda_{Pu} = \frac{A_{Pu}}{N_{Pu}} = \frac{4.4 \times 10^6}{5.0377 \times 10^{18}} = 8.73 \times 10^{-13} \text{ s}^{-1}$$

d. i. $$N_U = \frac{m_U}{M_U} \times N_A = \frac{6.0 \times 10^{-6}}{0.235 \text{ kg}} \times \left(6.02 \times 10^{23}\right) = 1.5370 \times 10^{19}$$

$$\text{fraction} = \frac{5.0377 \times 10^{18}}{5.0377 \times 10^{18} + 1.5370 \times 10^{19}} = 0.24685 = 0.25$$

ii. $$N_{Pu} = N_{Pu,0}e^{-\lambda t}$$

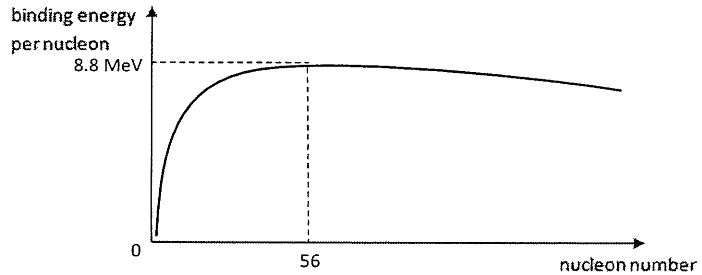
$$-\lambda t = \ln\frac{N_{Pu}}{N_{Pu,0}}$$

$$t = -\frac{1}{\lambda}\ln\frac{N_{Pu}}{N_{Pu,0}} = -\frac{1}{8.7342 \times 10^{-13}} \times \ln 0.24685 = 1.6017 \times 10^{12} \text{ s} = 5.08 \times 10^4 \text{ years}$$

14. [TJC/H2/Prelims 2010/P3/4]

a. i. See definitions in lecture notes.

ii.



iii. In nuclear fission, a heavy nucleus splits to give lighter daughter nuclei of greater binding energy per nucleon.

The difference in the total B.E. results in a large amount energy released during the process. This is a potential source of energy.

b. i. $$E = \Delta mc^2 = \left[\left(m_{^{235}_{92}U} + m_{^1_0 n}\right) - \left(m_{^{139}_{54}Xe} + m_{^{95}_{38}Sr} + 2m_{^1_0 n}\right)\right]c^2$$

$$= \left[m_{^{235}_{92}U} - m_{^{139}_{54}Xe} - m_{^{95}_{38}Sr} - m_{^1_0 n}\right]c^2$$

$$= [235.043929 \text{ u} - 138.918793 \text{ u} - 94.919359 \text{ u} - 1.008665 \text{ u}]c^2$$

$$= 2.94 \times 10^{-11} \text{ J}$$

ii. energy released from 2000 kg of uranium $= \left(2.944853\times10^{-11}\right)\times\dfrac{2000}{235.043929\times1.66\times10^{-27}} = 1.50951\times10^{17}$ J

useful energy $= \left(1.50951\times10^{17}\right)\times0.080 = 1.20761\times10^{16}$ J

time $= \dfrac{1.20761\times10^{16}}{1.15\times10^{17}} = 0.105010$ years $= 38.3$ days

15. [VJC/H2/Prelims 2010/P3/6]

a. i. See definitions in lecture notes.

ii. $\Delta m = 8\times1.007276\text{ u} + 8\times1.008665\text{ u} - 15.990527\text{ u} = 0.137001\text{ u}$

B.E. $= \Delta mc^2 = \left(0.137001\times1.66\times10^{-27}\right)\times\left(3.00\times10^8\right)^2 = 2.04679\times10^{-11}\text{ J} = 128\text{ MeV}$

iii. Binding energy per nucleon of ${}^{16}_{8}\text{O} = \dfrac{127.925}{16} = 8.00$ MeV

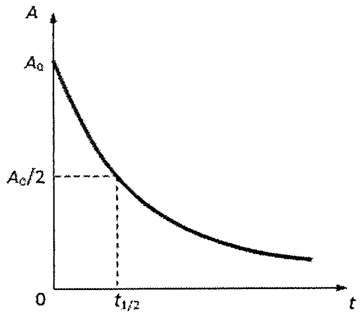
Binding energy per nucleon of ${}^{17}_{8}\text{O} = \dfrac{126.43}{17} = 7.44$ MeV

Since ${}^{16}_{8}\text{O}$ has a higher binding energy per nucleon, it would be more stable.

b. i. ${}^{40}_{19}\text{K} \rightarrow {}^{40}_{18}\text{Ar} + {}^{0}_{1}\beta$

ii. The probability that a particular potassium-40 nucleus will decay within a year is 5.3×10^{-10}.

iii.



iv. Amount of argon-40 nuclides: $n_{{}^{40}_{18}\text{Ar}} = \dfrac{0.84\times10^{-6}}{40} = 2.1\times10^{-8}$ mol

Amount of potassium-40 nuclides: $n_{{}^{40}_{19}\text{K}} = \dfrac{0.10\times10^{-6}}{40} = 2.5\times10^{-9}$ mol

$$n_{{}^{40}_{19}\text{K}} = n_{{}^{40}_{19}\text{K},0}\, e^{-\lambda t}$$

$$2.5\times10^{-9} = \left(2.5\times10^{-9} + 2.1\times10^{-8}\right)e^{-\lambda t}$$

$$t = -\frac{1}{5.3\times10^{-10}}\ln\left(\frac{2.5\times10^{-9}}{2.5\times10^{-9}+2.1\times10^{-8}}\right) = 4.23\times10^{9}\text{ years}$$

v. All the argon-40 is formed from the decay of potassium-40.

vi. $A = \lambda N = \left(\dfrac{5.3\times10^{-10}}{365\times24\times3600}\right)\times\left(2.5\times10^{-9}\right)\times\left(6.02\times10^{23}\right) = 0.0253$ Bq

Since the activity is quite low, simple safety precautions would suffice.

16. [YJC/H2/Prelims 2010/P3/8]

a. See definitions in lecture notes.

b. i. 36 protons and 56 neutrons.

ii. $\Delta m = 235.044\text{ u} + 1.009\text{ u} - 91.910\text{ u} - 140.916\text{ u} - 3\times(1.009\text{ u}) = 0.200\text{ u}$

$E = \Delta mc^2 = \left(0.200\times1.66\times10^{-27}\right)\times\left(3.00\times10^8\right)^2 = 2.99\times10^{-11}$ J

iii. $P_{\text{total}} = \dfrac{3.5\times10^{-3}}{235.044\times1.66\times10^{-27}}\times\left(2.988\times10^{-11}\right) = 2.6803\times10^{11}$ W

$P_{\text{useful}} = 0.23\times P_{\text{total}} = 0.23\times\left(2.6803\times10^{11}\right) = 6.16\times10^{10}$ W

iv. The energy is usually released as radiation and kinetic energy of product particles.

c. i. See definitions in lecture notes.

ii. Since the mass no of X is 234 – 230 = 4, it is most likely an alpha particle / a helium-4 nucleus.

iii. $\lambda = \dfrac{\ln 2}{t_{1/2}} = \dfrac{\ln 2}{2.4\times10^5} = 2.89\times10^{-6}\text{ years}^{-1}$

iv. $A = A_0e^{-\lambda t} = \lambda N_0 e^{-\lambda t} = \left(2.8881\times10^{-6}\right)\times\left(5.5\times10^{26}\right)\times e^{-\left(2.8881\times10^{-6}\right)\times\left(8.7\times10^4\right)} = 1.24\times10^{21}\text{ years}^{-1}$

v. Any two

- Used as radioactive tracers in either medicine, agriculture or pipe leakage
- Used to monitor thickness of papers or sheets produced in factory
- Used of ionising radiation in radiotherapy to treat cancer/tumour
- Used in smoke detectors to trigger alarm
- Used in carbon-dating to determine age of material